Supplementary Results — Module D

A structure-corrected two-locus screen for intrinsic incompatibilities

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This note reports the intrinsic (two-locus) arm. All results are structure-corrected but pre-simulation: the neutral, recombination-matched null (Methods, *Neutral, recombination-matched null*) is still required to convert the leads below into excess-over-neutral statements. Figures and tables are collected here.

1 Most associations reflected technical duplication or broad co-ancestry

Across 1.38×10^6 unlinked comparisons in the targeted scan, 276 passed Benjamini–Hochberg $q < 0.01$. Removing 29 likely duplicate or mismapped pairs, consolidating nearby correlated regions, and excluding associations driven by a single population left **110 associations among 83 regions** (18 negative and 92 positive; Table 1; Fig. 1).

Most of the remaining signal involved low-recombination regions: 66 of the 110 associations connected two such regions, whereas only 13 connected two regions outside this class. Four low-recombination regions had at least ten partners. Thus, the scan was dominated by a small number of broadly associated regions rather than by isolated candidate pairs.

2 Associations form several multi-chromosomal modules

Average-linkage clustering separated the network into 23 descriptive co-ancestry modules. The largest contained 16 regions on 12 chromosomes, followed by modules containing 14, 9, and 7 regions. Because these modules have distinct genotype patterns, they were examined separately rather than combined in one ordination. Hierarchical clustering of individuals within each module showed consistent genotype groupings across its member regions. In the illustrated module, this pattern extended across six chromosomes (Fig. 2). Overall, 79 of the 110 associations connected regions assigned to the same module (Fig. 3).

3 A small number of exploratory candidate associations remain

Thirteen associations connected two regions outside the low-recombination class (Table 2; Fig. 4). Most belonged to an *F. aquilonia* co-sorting module centred on F33454 (Chr10), whose partners occurred on seven other chromosomes and were predominantly sorted in the same ancestry direction. These positive associations are more naturally interpreted as coordinated ancestry sorting than as pair-specific incompatibilities. The strongest negative association joined F11431 and F49480 (Chr3–Chr16; $q = 2.2 \times 10^{-10}$). Neither endpoint was classified as repeatedly sorted, making this the highest-priority negative-association candidate. Several additional positive pairs joined regions sorted toward the same parental ancestry. The screen therefore produced a short exploratory candidate list rather than evidence for a large set of pairwise incompatibilities.

Table 1: Filtering of the structure-corrected targeted scan. Low recombination denotes a recombination percentile below 0.10 and is an annotation rather than an exclusion criterion.

Stage	Associations	Regions
Unlinked focal–partner tests	1,381,804	–
Benjamini–Hochberg $q < 0.01$	276	–
After removing likely duplicate mappings	247	156
After regional consolidation	160	127
After leave-one-population-out filtering	110	83
Final network	**110**	**83**
negative / positive	18 / 92	–
low–low / mixed / other–other	66 / 31 / 13	–
hubs (degree ≥ 10) / low-recombination regions	–	4 / 59

Table 2: The 13 associations for which both endpoints lie outside the low-recombination class, with chromosomes, association sign, false-discovery q, and ancestry-sorting class. The F33454 anchor and its aquilonia-sorted partners form a co-sorting module; neither endpoint of the strong negative F11431–F49480 association was repeatedly sorted.

Pair	Chr	association	q	sorting (a/b)
F11431–F49480	3/16	negative	2.2×10^{-10}	unsorted/unsorted
F10974–F33454	3/10	negative	2.7×10^{-4}	aqu/aqu
F32802–F33454	9/10	negative	1.1×10^{-3}	aqu/aqu
F11265–F33454	3/10	negative	6.2×10^{-3}	pol/aqu
F15419–F33454	4/10	positive	2.8×10^{-11}	aqu/aqu
F33454–F57178	10/19	positive	8.8×10^{-5}	aqu/aqu
F33454–F41301	10/13	positive	1.2×10^{-4}	aqu/aqu
F33454–F35575	10/11	positive	2.2×10^{-4}	aqu/aqu
F17114–F49429	4/15	positive	9.2×10^{-4}	pol/pol
F15820–F26284	4/7	positive	2.3×10^{-3}	pol/pol
F21099–F33454	5/10	positive	3.0×10^{-3}	aqu/aqu
F22294–F39587	6/12	positive	4.7×10^{-3}	aqu/aqu
F45183–F45504	14/14	positive	5.0×10^{-3}	unsorted/unsorted

4 Interpretive limits

Low recombination does not by itself imply either neutrality or selection. It can preserve long ancestry blocks, but it can also facilitate the maintenance of coadapted variation. The empirical scan therefore cannot determine whether the observed modules exceed neutral expectations. That question requires applying the same procedure to recombination-matched simulations. Until then, the principal result is that most apparent two-locus associations can be attributed to likely mapping duplication, broad co-ancestry, or single-population leverage, leaving a small and transparent exploratory candidate list.

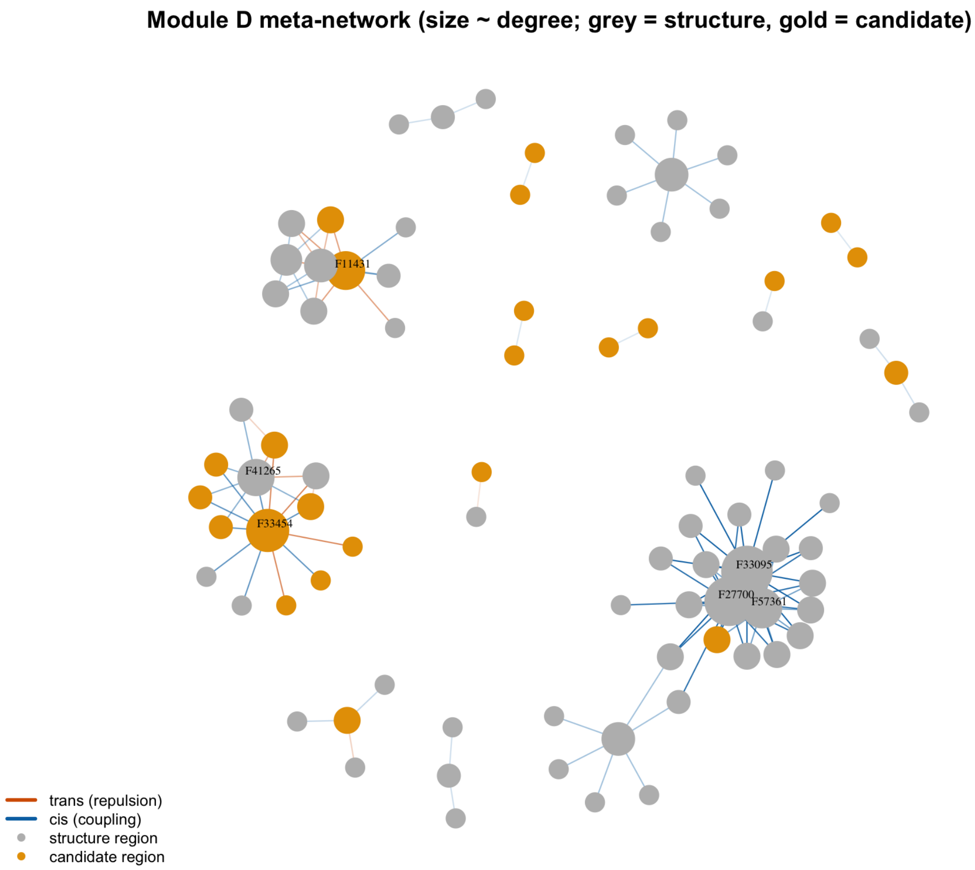

Figure 1: **The structure-corrected two-locus network.** Regions are sized by degree; grey denotes low recombination and gold denotes other regions. Blue edges are positive (coupling) and orange edges are negative (repulsion). The network contains a large low-recombination complex, the F33454 co-sorting module, and several isolated candidate pairs.

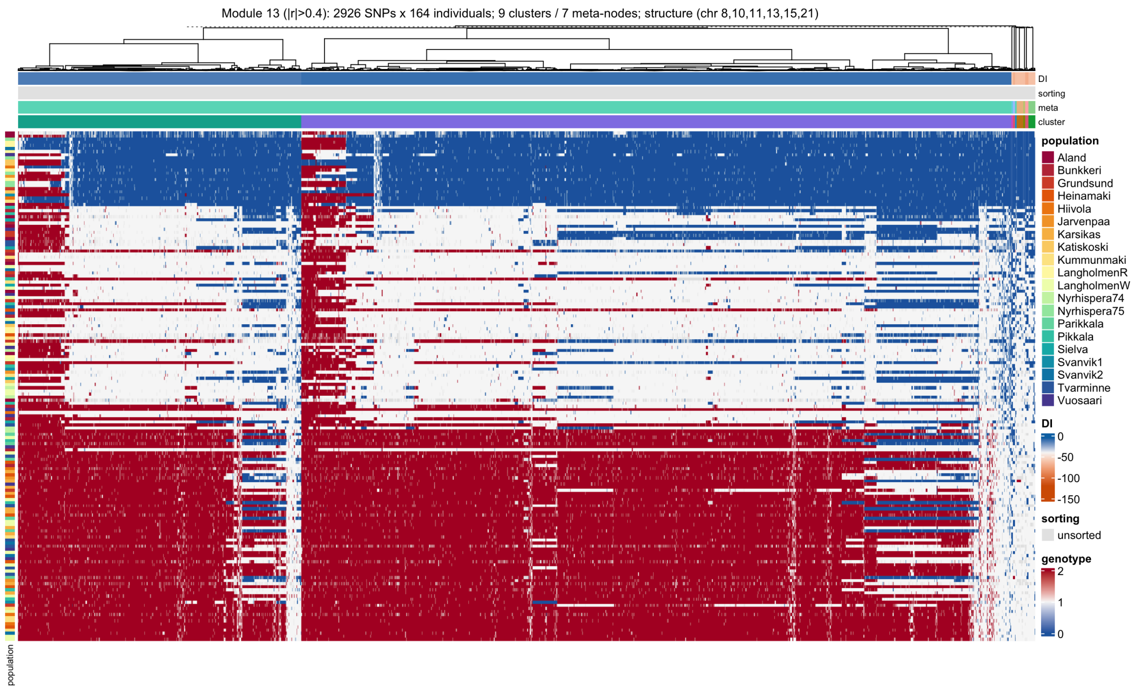

Figure 2: **A representative multi-chromosomal co-ancestry module.** Genotype heatmap of a module's member SNPs (9 clusters / 7 meta-nodes across Chr8/10/11/13/15/21; columns clustered; colour bars: cluster, meta-node, sorting, DI; rows hierarchically clustered by multilocus genotype). An individual is homozygous-reference (blue), heterozygous (pale), or homozygous-alternate (red) consistently across all 2,926 SNPs on all six chromosomes. The resulting groups do not coincide exactly with populations.

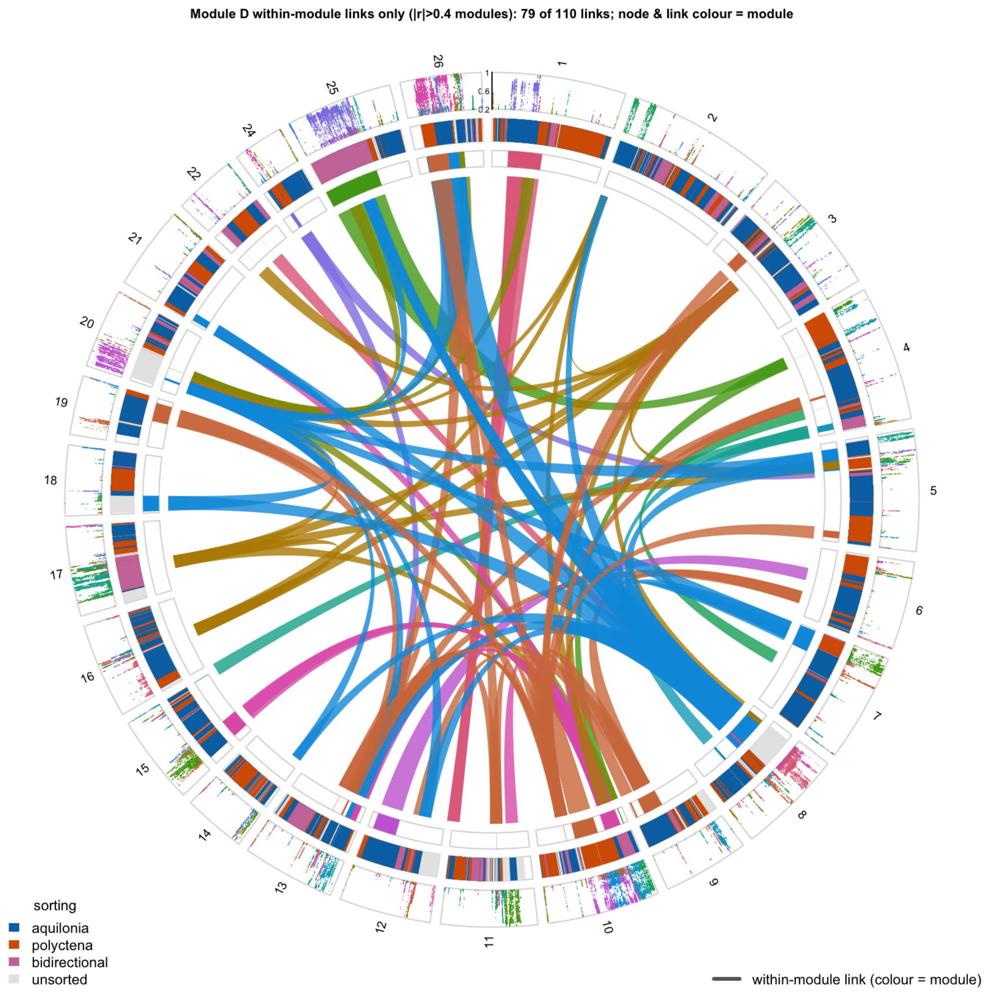

Figure 3: **Within-module associations.** Only edges whose two meta-nodes fall in the same cross-chromosome correlation module (79 of 110), coloured by module; node ring coloured to match. Each co-ancestry module reads as its own bundle of links across chromosomes. Outer track: local LD support ($ld_w > 0.2$) per marker; inner ring: the genome-wide sorting classification.

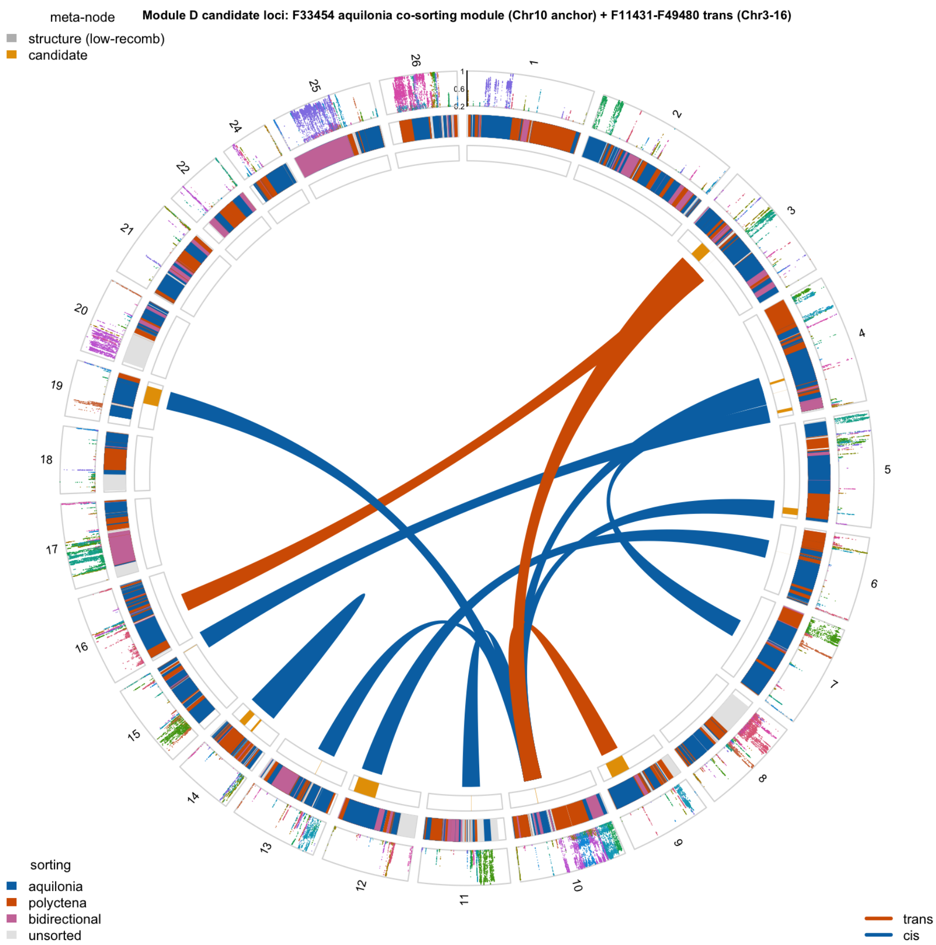

Figure 4: **Candidate associations on the genome.** Each association is drawn as a band spanning the two regions it joins (positive blue, negative orange; minimum width for visibility). The Chr10 anchor F33454 fans out to aquilonia-sorted partners on Chr3/4/5/9/11/13/19. Neither endpoint of the negative F11431–F49480 association (Chr3–Chr16) was classified as repeatedly sorted.